

Formulation Guide

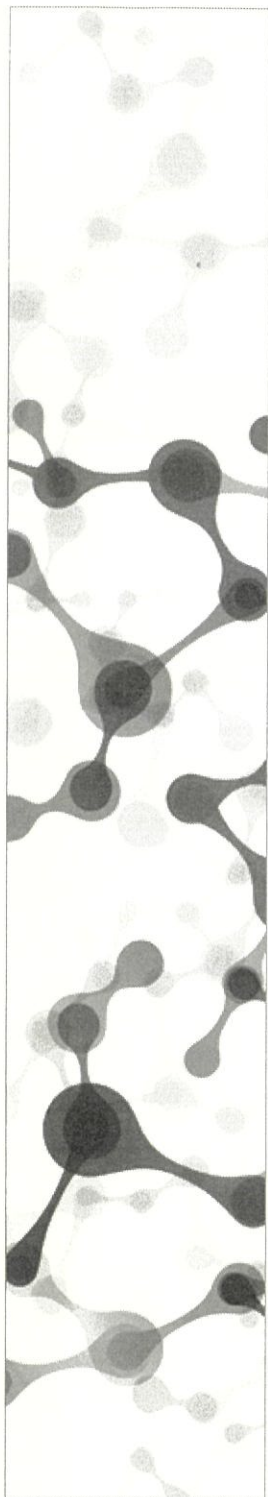
Sytenol® A

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SYTHEKON

...making innovation work

APPROVED DATE 28 AUG 2019



Sytenol® A (INCI: Bakuchiol)

Sytenol® A is the first natural alternative to Retinol without having any of the negatives associated with Retinol; Photochemically & hydrolytically stable; Can be used during the day; Multiple comparative studies revealed Sytenol A to be a true alternative to retinol. Modulates retinoid-binding & metabolizing genes; Significantly improves multiple dermal & dermo-epidermal junction genes/proteins; Collagen I, III & IV stimulator; Provides hydration by up-regulating synthesis of hyaluronic acid & aquaporin 3; Clinically proven to reduce multiple signs of aging - Significant reduction in roughness & dryness; fine lines & wrinkles and improvement in skin tone, elasticity & firmness; radiance and brightness. Sytenol® A is a natural and well-defined material with a purity level of $\geq 90\%$.

Formulation Guidelines

Formulated product containing Sytenol® A will be white and remain stable over time. Long term stability testing on emulsions containing 1% Sytenol® A showed practically no color shift at 45°C and RT.

- Sytenol® A should be used at a level of 0.5 – 1% (w/w) of finished formulation.
- Sytenol® A is miscible in a wide variety of emollients (for example, capric/caprylic triglycerides, C12-15 alkyl benzoates, dicaprylyl ether), vegetable oils (sunflower oil, jojoba oils, olive oil) and silicones (dimethicones, cyclomethicones).
- Dissolve Sytenol® A in appropriate hydrophobic emollients and add to the formulation after emulsification. Solution of Sytenol® A may be added to the formulation with a processing temperature of about 50°C or below. Alternately, Sytenol® A can be added directly to the oil phase. Avoid prolonged heating of the oil phase above 75°C.
- For preparation of anhydrous serum or gel, use silicone derivatives such as, Dimethicone, Cyclomethicone, and Dimethiconol & Dimethicone Crosspolymer.
- Sytenol® A is a phenolic compound. Therefore, presence of iron or copper ions must be avoided to eliminate coloration due to the interaction with these metal ions. Addition of a small amount of disodium EDTA (0.1%) or other chelating agents resolves this problem.
- The finished product must be acidic, preferably having pH below 6.5. A pH below 5.5 is recommended when potassium sorbate or sodium benzoate are used as preservatives.
- Avoid prolong exposure to sunlight or heat. Opaque and airless packaging is highly recommended for maintaining product integrity and stability over time.

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Anti-Aging Night Cream with Sytenol® A and HydraSynol™ DOI

Formulation#RC01-13

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Water	Water (demineralized)	QS
Disodium EDTA	Versene Na/Dow	0.10
Glycerin	Emery 916/BASF	3.00
Butylene Glycol	Butylene Glycol/Ruger	2.00
D Panthenol	Panthenol/BASF	0.50
Phase A2		
Xanthan Gum	Keltrol® CG-T/CP Kelco	0.15
Carbomer	Carbopol Ultrez 10/Lubrizol	0.40
Sucrose Stearate	Sisterna SP50C/MMP	2.50
Poly C8-22 Alkyl Acrylate/Metacrylic Acid Cross Polymer	Intelimer 8600/Air Products	2.50
Phase B		
Simmondsia Chinensis (Jojoba Oil)	Jojoba Oil/Jeen Intl.	3.00
Shea Butter	Shebu Refined/Rita	2.50
Behenyl Alcohol	Lanette 22/BASF	1.50
Potassium Cetyl Phosphate	Amphisol® K/DSM	1.00
Stearic Acid	Pristerene® 4911/Croda	1.00
Dimethicone	Dow Corning 200, 100cst/Dow Corning	4.00
Natural Beeswax	Beeswax/Rita	2.00
Isosorbide Dicaprylate	HydraSynol™ DOI/Sytheon	3.00
Alpha Tocopherol	Vitamin E/BASF	0.20
Hydrogenated Polyisobutene	Panalane L14E/Lipo	4.00
Bakuchiol	Sytenol® A/Sytheon	0.50
Phase C		
Sodium Hydroxide	Sodium Hydroxide (50%sol)	0.30
Phase D		
Fragrance	Women's FragranceK3258/Carruba	0.20
Red 33 (CI 17200)	D&C Red 33 (1% sol)	0.05
Phenoxyethanol, Ethylhexylglycerine	Euxyl PE 9010/Schülke	1.00
Total		100.00

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Procedure:

Combine A1; disperse A2 one by one in A1 while stirring and heat A to 75°C. Combine ingredients of phase B and heat to 75°C. Add phase B to A with good mixing. Homogenize mixture at moderate speed for 3-5 min while adding phase C (adjust pH to 5.5-6.0). Cool batch to 40°C with propeller agitation until mixture is homogeneous. Add phase D ingredients while continue mixing.

Notes:

pH value: 5.5-6.0; Viscosity: 80,000-100,000mPas (Brookfield RVT, Spindle C, 10rpm) at 25°C

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Anti-Aging Repair Serum with Sytenol® A and HydraSynol™ DOI

Formulation#RC01-34

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Water	Water (demineralized)	QS
Disodium EDTA	Versene Na/Dow	0.10
Glycerin	Emery 916/BASF	3.00
Butylene Glycol	Butylene Glycol/Ruger	2.00
D Panthenol	Panthenol/BASF	0.20
Phase A2		
Acrylates/C10-30 Alkyl Acrylate Crosspolymer	Carbopol Ultrez 21/Lubrizol	0.15
Phase B		
Triethanolamine	Triethanolamine 99%/Ruger	0.10
Phase C		
Isosorbide Dicaprylate	HydraSynol™ DOI/Sytheon	2.00
Isostearyl Alcohol, Butylene Glycol Cocoate, Ethylcellulose	Emulfree CBG/Gattefosse	2.00
Cyclopentasiloxane, Dimethiconol	Dow Corning 1501 Fluid/Dow Corning	2.50
Bakuchiol	Sytenol® A/Sytheon	0.50
Phase D		
Hydroxyethylacrylate/sodium Acryloyldimethyltaurate Copolymer, Squalane, Polysorbate 60	Simulgel NS/Seppic	1.50
Phase E		
Poly C8-22 Alkyl Acrylate/Metacrylic Acid Cross Polymer	Intelimer 8600/Air Products	2.00
Phase F		
Fragrance	Green Citrus#61044049/Bell Flavors	0.10
Phase G		
Phenoxyethanol, Ethylhexylglycerine	Euxyl PE 9010/Schülke	1.00
Total		100.00

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Procedure:

Combine A1; disperse A2 one by one in A1 while stirring and heat A to 40°C. Combine ingredients of phase C and heat to 40°C. Neutralize phase A with phase B to pH 5.5. Add phase C to AB with good mixing. Homogenize mixture at moderate speed for 3-5 min while adding phase D. Switch batch to propeller agitation until mixture is homogeneous. Add phase E, F & G while continue mixing.

Notes:

pH value: 5.5-6.0; Viscosity: 25,000-35,000mPas (Brookfield RVT, Spindle C, 10rpm) at 25°C

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Age-Defying Day Cream with Sytenol® A and HydraSynol™ DOI

Formulation#RC01-42

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Water	Water (demineralized)	QS
Disodium EDTA	Versene Na/Dow	0.10
Glycerin	Emery 916/BASF	2.00
Butylene Glycol	Butylene Glycol/Ruger	2.00
D Panthenol	Panthenol/BASF	0.50
Phase A2		
Xanthan Gum	Keltrol® CG-T/CP Kelco	0.10
Acrylates/C10-30 Alkyl Acrylate Crosspolymer	Carbopol Ultrez 21/Lubrizol	0.30
Sucrose Stearate	Sistema SP50C/MMP	1.50
Phase B		
Simmondsia Chinensis (Jojoba Oil)	Jojoba Oil/Jeen Intl.	3.00
Shea Butter	Shebu Refined/Rita	1.50
Behenyl Alcohol	Lanette 22/BASF	1.25
Potassium Cetyl Phosphate	Amphisol® K/DSM	1.00
Stearic Acid	Pristerene® 4911/Croda	1.00
Dimethicone	Dow Corning 200,50 cst/Dow Corning	5.00
Natural Beeswax	Beeswax/Rita	1.50
Isosorbide Dicaprylate	HydraSynol™ DOI/Sytheon	2.00
Alpha Tocopherol	Vitamin E/BASF	0.20
Ammonium Acryloyldimethyltaurate/VP Copolymer	Aristoflex AVC/Clariant	0.30
Phase C		
Sodium Hydroxide	Sodium Hydroxide (50%sol)	0.20
Phase D		
Bakuchiol	Sytenol® A/Sytheon	1.00
Phase E		
Phenoxyethanol, Ethylhexylglycerine	Euxyl PE 9010/Schülke	1.00
Phase F		
Fragrance	Women's Fragrance K3258/Carrubba	0.20
Total		100.00

Procedure:

Combine A1; disperse A2 one by one in A1 while stirring and heat A to 75°C. Combine ingredients of phase B and heat to 75°C. Add phase B to A with good mixing. Homogenize mixture at moderate speed for 3-5 min while adding phase C (adjust pH to 5.5-6.0). Start cooling batch while adding phase D. At 40°C, add phase E & F; mix with propeller agitation until mixture is homogeneous.

Notes:

pH value: 5.5-6.0; Viscosity: 80,000-100,000mPas (Brookfield RVT, Spindle C, 10rpm) at 25°C

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Acne-Control Lotion with Sytenol® A and Salicylic Acid

Formulation#AC-RC01-11

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Water	Water (demineralized)	QS
Disodium EDTA	Versene Na/Dow	0.10
Glycerin	Glycerin 99%/Ruger	3.00
Butylene Glycol	Butylene Glycol/Ruger	2.00
Salicylic Acid	Salicylic Acid/Universal Preserv	2.00
Phase A2		
Xanthan Gum	Vanzan NF/Vanderbilt	0.15
Acrylates/C10-30 Alkyl Acrylate Copolymer	Carbopol Ultrez 21/Lubrizol	0.15
Phase B		
Sodium Hydroxide (50%)	Sodium Hydroxide (to pH 4.5)	5.00
Phase C		
Dicaprylyl Ether	Cetiol OE/BASF	4.00
Cetyl Alcohol	Crodacol C-70/Croda	1.50
Tocopheryl Acetate	Vitamin E Acetate/DSM	0.20
Glyceryl Stearate, PEG-100 Stearate	Arlacel 165/Croda	2.00
Cetearyl Alcohol, Ceteareth-20	Promulgen D/Lubrizol	1.00
Dimethicone	DC, 200/50CST/Dow Corning	1.00
Cyclomethicone, Polysilicone-11	Gransil GCM-5/Grant Industries	5.00
Beeswax	Natural Beeswax/Rita Corp	1.00
Bakuchiol	Sytenol® A/Sytheon	1.00
Phase D		
Polyacrylamide, C13-14 Isoparaffin, Laureth-4	Sepigel 305/Seppic	1.50
Phase E		
Phenoxyethanol, Ethylhexylglycerine	Euxyl PE 9010/Schülke	1.00
Total		100.00

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Procedure:

Combine A1; disperse A2 in A1 while stirring and heat A to 75°C. Combine phase B ingredients then heat to 75°C. Add phase B to A with good mixing. Homogenize mixture at high speed, while adding phase C & D. Cool batch to 45°C; Add phase E; stir gently until mixture is homogeneous.

Notes:

pH value: 4.50; Viscosity: 25,000-45,000 cps (Brookfield RVT, spindle C, Helipath 10rpm) at 25°C

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Anti-Aging Hydrating Gel Cream with Sytenol® A and HydraSynol™ DOI

Formulation#RC01-56

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Water	Water (demineralized)	QS
Disodium EDTA	Versene Na/Dow	0.10
Hydrogenated Lecithin	Emulmetik 320/Lucas Meyer	0.75
Glycerin	Glycerine 99%/Ruger	2.00
PEG-8	Pluracare E 400/BASF	2.00
Niacinamide	Niacinamide PC/DSM	2.00
Caffeine	Caffeine Anhydrous USP/MMP International	0.50
Butylene Glycol	Butylene Glycol/Ruger	3.00
Phase A2		
Ammonium acryloyldimethylsulfate/VP copolymer	Aristoflex AVC/Clariant	1.10
Phase B		
Cyclopentasiloxane	DC 345/Dow Corning	5.00
Dimethicone	DC 200,50cst/Dow Corning	1.00
Tocopheryl Acetate	Vitamin E Acetate/DSM	0.10
Isosorbide Dicaprylate	HydraSynol™ DOI/Sytheon	2.00
Oleth-10	Brij O10/Croda	0.50
Bakuchiol	Sytenol® A/Sytheon	0.50
Phase D		
Citrus Aurantium Bergamia (Bergamot) Fruit Oil	Bergamont Oil/Premier	0.15
Phase E		
Phenoxyethanol, Ethylhexylglycerine	Euxyl PE 9010/Schuelke	1.00
Total		100.00

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Procedure:

Combine A1 and heat to 65°C. Disperse A2 in A1 with rapid mixing. Combine phase B ingredients and heat to 65°C. Add phase B to A1-A2 with good mixing. Homogenize mixture at high speed for 2 min. Cool batch to 45°C; Add phase D and E; stir gently until mixture is homogeneous.

Notes:

pH value: 6.00; Viscosity: 35,000-50,000 cps (Brookfield RVT, spindle C, Helipath 10rpm) at 25°C

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Anti-Aging Illuminating Cream with Sytenol® A and Synovea® HR

Formulation#RC01-76

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Water	Water (demineralized)	QS
Disodium EDTA	Versene Na/Dow	0.10
Glycerin	Glycerine 99%/Ruger	2.00
Niacinamide	Niacinamide PC/DSM	1.00
Butylene Glycol	Butylene Glycol/Ruger	3.00
Panthenol	Ritapan DL 50%/Rita	0.20
Phase A2		
Ammonium Acryloyldimethyltaurate/VP Copolymer	Aristoflex AVC/Clariant	0.60
Phase B		
Rice Bran Oil	Lipovol RB/Lipo	3.50
Cyclopentasiloxane, Petrolatum, Polysilicone 11	Gransil PS-5/Grant Industries	6.00
Butyrospermum Parkii (Shea Butter)	Shebu Refined/Rita	1.00
Tocopheryl Acetate	Vitamin E Acetate/DSM	0.20
Glyceryl Stearate, PEG-100 Stearate	Arlacel 165/Croda	2.2.0
Behenyl Alcohol	Lanette 22/BASF	1.10
Dimethicone	DC, 200/100CST/Dow Corning	3.00
Isosorbide Dicaprylate	HydraSynol™ DOI/Sytheon	2.00
Ethyl Linoleate	Synovea® EL/Sytheon	0.50
Phase C		
Backuchiol	Sytenol® A/Sytheon	1.00
Phase D		
Polyacrylamide, C13-14 Isoparaffin, Laureth-4	Sepigel 305/Seppic	1.00
Phase E		
Hexylresorcinol	Synovea® HR/Sytheon	1.00
PEG-8	Pluracare E-400/BASF	2.00
Phase F		
Phenoxyethanol, Ethylhexylglycerine	Euxyl PE 9010/Schülke	1.00
Bergamont Oil	Bergamont Oil/Premier	0.15
Total		100.00

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Procedure:

Combine A1; disperse A2 one by one in A1 while stirring and heat A to 75°C. Combine ingredients of phase B and heat to 75°C. Add phase B to A with good mixing. Homogenize mixture at moderate speed for 3-5.min. Add phase C at 60°C followed by Phase D. Cool batch to 40°C with propeller agitation until mixture is homogeneous. Add phase E & F while continue mixing.

Notes:

pH value: 5.5-6.0; Viscosity: 40,000-60,000mPas (Brookfield RVT, Spindle C, 10rpm) at 25°C

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Anti-Aging and Blemish Control Gel with Sytenol[®] A

Formulation#RC01-130

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Water	Water (demineralized)	QS
Disodium EDTA	Versene Na/Dow	0.10
Polyquaternium-10	UCARE [™] Polymer LR 400/Dow	0.50
Phase A2		
PEG-8	Pluracare E 400/BASF	10.00
Panthenol	D-Panthenol/DSM	0.50
1,3 Propanediol	Zemea/DuPont	8.00
Phase B		
Glycolic Acid	DuPont [™] Glypure [®] 70/DuPont	5.00
Sodium Hydroxide	Sodium Hydroxide 50% soln	0.50
Phase C		
Alcohol	SD Alcohol 40 B	35.00
Salicylic Acid	Salicylic Acid	1.50
Phase D		
Backuchiol	Sytenol [®] A/Sytheon	0.50
PEG-40 Hydrogenated Castor Oil	Lipocol HCO-40/Lipo Chemicals	0.75
Total		100.00

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Procedure:

Combine A1; add ingredients of phase A2 one by one and heat to 40° C. Premix phase B and heat and add to A. Add phase C & D to AB with good mixing. A clear translucent serum should be obtained.

Notes:

pH value: 3.75-4.0

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Overnight Renewal Face Oil with Sytenol® A, HydraSynol™ DOI and Synovea® EL

Formulation#RC02-26

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Caprylic/Capric Triglycerides	Myritol 318/BASF	QS
Simmondsia Chinensis (Jojoba Oil)	Jojoba Oil/Jeen Intl.	15.00
Dicaprylyl Ether	Cetiol OE/BASF	40.00
Isosorbide Dicaprylate	HydraSynol™ DOI/Sytheon	2.00
Prunus Amygdalus Dulcis (Sweet Almond) Oil	Cropure Almond/Croda	10.00
Tocopheryl Acetate	Vitamin E Acetate/BASF	0.20
Ethyl Linoleate	Synovea® EL/Sytheon	1.00
Moringa Oleifera Seed Oil	Floralipids Moringa Oil/Floratech	2.50
Bisabolol	RonaCare Bisabolol/EMD Chemicals	0.30
Citrus Aurantium Bergamia (Bergamot) Fruit Oil	Bergamot Oil/Premier Specialties	0.15
Lavender (Lavendula Angustifolia) Oil	Lavender Oil/Premier Specialties	0.20
Pelargonium Graveolens Leaf Oil	Geranium Oil/Premier Specialties	0.02
Phase B		
Bakuchiol	Sytenol® A/Sytheon	0.50
Total		100.00

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Procedure: Prepare phase A and mix well at room temperature. Add phase B and mix until uniform.

Hydrating Anti-Aging Oil with Sytenol® A, HydraSynol™ DOI and HydraSynol™ IDL

Formulation#RC02-32

INCI name	Trade Name/Supplier	% w/w
Prunus Armeniaca (Apricot) Kernel Oil	Apricot Kernel Oil/Jedwards International	36.00
Caprylic/Capric Triglycerides	Myritol 318/BASF	25.00
Ethyl Macadamiate	Floramac 10/Floratech	8.75
Helianthus Annuus (Sunflower) Seed Oil	FloraSun 90/Floratech	3.00
Isosorbide Dicaprylate	HydraSynol™ DOI/Sytheon	2.00
Isosorbide Disunflowerseedate	HydraSynol™ IDL/Sytheon	2.00
Tocopheryl Acetate	Vitamin E Acetate/BASF	0.20
Persea Gratissima (Avocado) Oil	Avocado Oil/Jedwards International	1.00
Dicaprylyl Ether	Cetiol OE/BASF	20.00
Squalane	Fitoderm/Centerchem	1.00
Bisabolol	RonaCare Bisabolol/EMD Chemicals	0.20
Lavender (Lavendula Angustifolia) Oil	Lavender Oil/Premier Specialties	0.10
Citrus Nobilis (Mandarin Orange) Peel Oil	Mandarin Oil/Ungerer	0.15
Citrus Paradisi (Grapefruit) Peel Oil	Grapefruit Oil/Premier	0.10
Bakuchiol	Sytenol® A/Sytheon	0.50
Total		100.00

Procedure: Mix well all ingredients at room temperature.

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